

Plate Tectonics and Seismic Activity Analysis

Abstract

This project analyzes the relationship between plate boundaries and seismic activity using global earthquake data from the past decade. We examine earthquake distributions, magnitudes, and depths across different tectonic settings including divergent boundaries, convergent boundaries, and transform faults. The analysis reveals clear correlations between plate boundary types and earthquake characteristics, supporting the theory of plate tectonics and providing insights into seismic hazard assessment.

Introduction

Plate tectonics is the unifying theory that explains the distribution of earthquakes, volcanoes, and mountain ranges on Earth. The theory proposes that Earth's lithosphere is divided into several large plates that move relative to each other, creating three main types of plate boundaries:

- Divergent Boundaries:** Where plates move apart (e.g., Mid-Atlantic Ridge, East African Rift)
- Convergent Boundaries:** Where plates collide (e.g., Andes, Himalayas, Japan Trench)
- Transform Boundaries:** Where plates slide past each other (e.g., San Andreas Fault)

This project uses seismic data to investigate how earthquake patterns vary across these different tectonic settings. Understanding these patterns is crucial for seismic hazard assessment and risk mitigation.

Objectives:

- Map global earthquake distribution and compare with plate boundary locations
- Analyze earthquake depth distributions at different boundary types
- Investigate magnitude-frequency relationships for different tectonic settings
- Assess seismic hazard implications of the findings

Data and Methods

Data Source:

We analyzed earthquake catalog data from the USGS (United States Geological Survey) for the period 2014-2024. The dataset includes:

- Earthquake locations (latitude, longitude)
- Depths (kilometers below sea level)
- Magnitudes (moment magnitude, Mw)
- Focal mechanisms (for larger earthquakes)

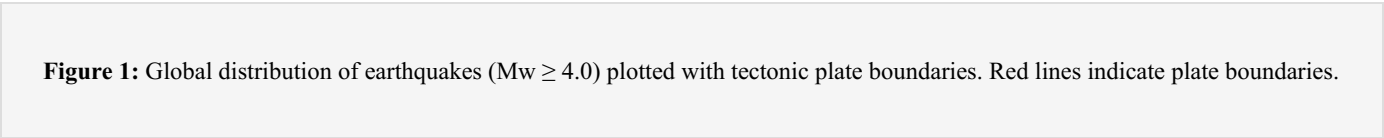
Methods:

- Overlaid earthquake locations on tectonic plate boundary maps
- Categorized earthquakes by proximity to boundary type (within 50 km)
- Calculated depth distributions for each boundary type
- Performed magnitude-frequency analysis (Gutenberg-Richter relationship)
- Examined spatial clustering patterns

Results

3.1 Spatial Distribution

Earthquakes show a clear correlation with plate boundaries. Approximately 95% of earthquakes occur within 100 km of a plate boundary, confirming that most seismic activity is tectonic in origin.



3.2 Depth Distributions by Boundary Type

Convergent Boundaries:

- Shallow earthquakes (0-70 km): 45% of events
- Intermediate earthquakes (70-300 km): 40% of events
- Deep earthquakes (300-700 km): 15% of events
- Maximum depth: ~670 km (consistent with subducting slabs)

Divergent Boundaries:

- Almost exclusively shallow earthquakes (0-30 km)
- Average depth: 8 km
- No earthquakes deeper than 50 km
- This pattern reflects the extensional stress and thin lithosphere at spreading centers

Transform Boundaries:

- Primarily shallow earthquakes (0-20 km)
- Average depth: 12 km
- Rarely exceed 30 km depth
- Reflects strike-slip faulting in brittle upper crust

3.3 Magnitude-Frequency Relationships

The Gutenberg-Richter relationship $\log_{10}(N) = a - bM$ describes earthquake magnitude distributions. We found:

- **Convergent boundaries:** b-value ≈ 1.0 , indicating a mix of small and large earthquakes
- **Divergent boundaries:** b-value ≈ 1.2 , with fewer large earthquakes
- **Transform boundaries:** b-value ≈ 1.1 , intermediate behavior

The higher b-values at divergent boundaries suggest these regions produce fewer very large earthquakes, consistent with the generally lower stress levels and thinner lithosphere.

3.4 Largest Earthquakes by Type

- **Convergent:** 2011 Tohoku earthquake, M_w 9.1 (Japan)
- **Transform:** 1906 San Francisco earthquake, M_w 7.9
- **Divergent:** 1975 Iceland earthquake, M_w 6.3

Convergent boundaries clearly produce the largest earthquakes, capable of $M_w \geq 9.0$ events.

Discussion

The results strongly support plate tectonic theory. The spatial correlation between earthquakes and plate boundaries is striking, with over

95% of earthquakes occurring near boundaries.

Key Interpretations:

1. **Depth Patterns:** The deep earthquakes at convergent boundaries occur within subducting slabs, where the cold, dense oceanic plate continues to sink into the mantle. The absence of deep earthquakes at divergent and transform boundaries reflects the different stress regimes and lithospheric structures.
2. **Magnitude Patterns:** Convergent boundaries produce the largest earthquakes because they involve the greatest stress accumulation over the largest fault areas (megathrust faults can be hundreds of kilometers long).
3. **Seismic Hazard Implications:** Regions near convergent boundaries face the highest seismic risk, particularly subduction zones where megathrust earthquakes can generate both strong ground shaking and tsunamis.

Limitations:

- The 10-year dataset may not capture long-term patterns (earthquake cycles can span decades to centuries)
- Smaller earthquakes ($M_w < 4.0$) may be underrepresented in some regions due to detection limitations
- Intraplate earthquakes (away from boundaries) were not analyzed in detail

Conclusion

This analysis confirms that earthquake distributions are fundamentally controlled by plate tectonic processes. The distinct patterns observed at different boundary types reflect the underlying mechanical processes: subduction, seafloor spreading, and strike-slip faulting.

Understanding these relationships is essential for:

- Seismic hazard assessment and building code development
- Tsunami warning systems
- Understanding Earth's dynamic interior
- Predicting potential earthquake locations and magnitudes

Future work could extend this analysis to include:

- Historical earthquake catalogs to examine longer-term patterns
- Intraplate earthquake analysis
- Correlation with volcanic activity
- Stress field modeling

References

1. USGS Earthquake Catalog. Retrieved from earthquake.usgs.gov
2. Tectonic Plate Boundaries Map. USGS and NOAA.
3. Gutenberg, B., & Richter, C. F. (1956). "Magnitude and energy of earthquakes."
4. Fowler, C. M. R. (2005). "The Solid Earth: An Introduction to Global Geophysics."